

Watermarking-based Defense Mechanism in LFC of Electricity Grids Compromised by Covert Attacks

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Abstract—This paper focuses on improving the load frequency control (LFC) performance of a multi-area power system compromised by covert attacks. In this regard, a novel detection technique based on the multiplicative watermarking and Luenberger observer is developed to identify covert attacks applied to the system. In the presented detection strategy, a special watermark generator filter produces a dummy signal on the original signal transmitted through the insecure communication channels. In addition, there is a watermark equalization filter at the end of the communication channel, which cancels out the impact of the watermark signal, resulting in the recovery of the original signal. A PI controller tuned by the deep reinforcement learning algorithm and dependent on the Luenberger observer is utilized as the secondary frequency control strategy to mitigate the impacts of the covert attacks. The OPAL-RT real-time emulator-based results show the proposed defense strategy successfully detects and mitigates the covert attacks imposed on the system.

Index Terms—Load frequency control, covert attacks, multiplicative watermarking, deep reinforcement learning, Luenberger observer, multi-area power system.

I. INTRODUCTION

A Variety of closed-loop control systems, such as load frequency control (LFC) with geographically distributed sensing and actuation components, is employed to operate power systems [1]. Wide-area monitoring system-based applications are used to obtain the system's real-time data under different conditions [2], [3]. Automatic generation control (AGC) service, as the most important section of the LFC application, controls the frequency and tie-line power of electricity grids using remote data transmitted by communication networks. Due to the presence of the AGC service in LFC schemes, power systems are more susceptible to cyberattacks, compromising the system's stability [4], [5].

In recent years, cyber-security has become an increasingly important topic for LFC systems, and a number of literature have been accomplished to study the implications and design countermeasures regarding different cyberattacks, including denial-of-service [5], and false data injection [6], [7] attacks. In [8], a survey on learning-based methods has been presented to detect cyberattacks in communication network-based systems. The defense mechanisms presented in the literature are not

effective versus stealth cyberattacks. Thus, more research is needed on stealthy attacks in LFC systems. The covert attack is a stealth attack strategy that can severely compromise the system's performance without detection. Accordingly, this paper studies the LFC performance of power systems compromised by covert threats and proposes a defense mechanism against these attacks.

Despite the presence of detection logic in the energy management systems of the LFC application, various stealthy attacks with comprehensive plant dynamics knowledge have remained undetected [9]. The covert attack is regarded as one of those stealthy attacks capable of disruption and disclosure with a high level of model knowledge. When an attacker generates control actions in the covert attack setup, the controller does not detect abnormalities because the covert agent removes the plant actuation effect by subtracting output measurements from output responses. Moreover, the adversary utilizes retained thresholds to reduce the number of false alarms caused by disturbances and model uncertainties [10]. As a result, despite the discrepancies between the attacker's plant model and the real system, the covert adversary remains a hidden cybernetic ally. A better description of the attack would be that attackers first try to interfere with the network channel between the controller and actuator to disrupt the power system's performance. After that, some carefully planned signals are applied to the sensor or controller channel to eliminate input attack effects [11]. In this way, even though the power system may have been compromised, the measurement signals remain the same, enabling these covert attacks undetected by the normal detection techniques described above.

To date, only a few techniques have been developed to identify covert attacks. In [12], the authors proposed a detection scheme consisting of model-based distributed estimators to identify local covert attacks in linear interconnected systems. A two-module distributed monitoring architecture comprising a set of unknown input observers was proposed in [13] to detect covert attacks applied to a DC microgrid considering the ideality of the communication channel. An unsupervised machine learning-based strategy named *i*-Forest was presented in [14] to identify covert attacks imposed on communication networks of power systems. The authors in [15] mentioned that data-driven-based techniques are not useful when the bogus measurement data vary slightly from learned standard use patterns. The authors in [16] proposed an intrusion detection scheme based on the GPS data and offline maps for intelligent transportation systems compromised by covert attacks. In [11], a recursive watermarking-based scheme was proposed to iden-

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tify covert attacks in an industrial cyber-physical system. The available identification strategies exhibit drawbacks, including performance failure and identification delay.

Regarding the inability of available methods to encounter stealth cyberattacks and the importance of preserving the grid's frequency in a permissible range under these attacks, this paper investigates the LFC performance of an interconnected power grid under stealth covert attacks. In this regard, covert attacks are designed on a two-area LFC system. Moreover, a defense mechanism based on dynamic multiplicative watermarking technique is proposed to encounter covert attacks. In the proposed defense mechanism, the multiplicative watermarking technique, a Luenberger observer, and a detector are employed to identify covert attacks. The multiplicative watermarking superimposes a watermark signal on an original signal before transmitting it over an insecure network. A reverse watermark is also used after the watermark signal transits via an insecure network, which mitigates the watermark's effect under any conditions. The motivation for containing the proposed watermarking technique is to make covert attacks identifiable by having them cause a defective reconstruction of the plant's signals. Unlike all watermarking-based methods, the multiplicative watermarking-based methods generally have a higher capacity for embedding information than other methods. Moreover, adversaries in this method cannot identify or copy the watermark parameters for estimating or removing the watermark signal and compromising the signal integrity. A Luenberger observer is also designed in the proposed mechanism to provide the system's state estimation and residual signal. A deep reinforcement learning (DRL)-based PI controller is also used in the AGC loop for frequency regulation of the system compromised by covert attacks. The proposed defense mechanism has the capability of coping with other stealth cyberattacks and can be applied to different real-world applications.

II. STUDIED LFC SYSTEM MODEL

Fig. 1 depicts the block diagram of the i^{th} control area, comprised of the thermal generation unit for a G -area LFC system. Studies accomplished on the LFC issues have widely used linearized models to analyze power system frequency behavior. The linearized LFC system model consists of a steam turbine, governor, rotating mass, and an AGC loop, as illustrated in Fig. 1. The mathematical model of the i^{th} control area can be expressed by the following:

$$\begin{cases} \dot{x}_i(t) = A_i x_i(t) + B_i u_i(t) \\ y_i(t) = C_i x_i(t) \end{cases} \quad (1)$$

In this model, A_i , B_i , and C_i are the i^{th} control area state-space matrices. $x_i(t) = [\Delta P_{tie,i} \ \Delta f_i \ \Delta P_{g,i} \ \Delta P_{t,i}]^T$ is the vector of the state variables in the i^{th} control area. $y_i(t) = [\Delta P_{tie,i} \ \Delta f_i]^T$ shows the vector of the i^{th} control area output variables, and $u_i(t)$ is the vector of the i^{th} area control signal. The process and measurement noises are also considered in modeling the power system dynamic as distributed Gaussian random variables with zero mean and covariance matrices $Cov(w_i) = \text{diag}[1.5e^{-5} \ 0 \ 0 \ 0]$ and $Cov(v_i) = [6e^{-6} I_2]$,

respectively. The mathematical equations related to the i^{th} control area state variables are written as follows:

$$\begin{cases} \Delta \dot{P}_{tie,i} = 2\pi \sum_{j=1, i \neq j}^G T_{ij} (\Delta f_i - \Delta f_j) \\ \Delta \dot{f}_i = \frac{1}{M_i} (\Delta P_{t,i} - \Delta P_{L,i} - \Delta P_{tie,i} - D_i \Delta f_i) \\ \Delta \dot{P}_{g,i} = \frac{1}{T_{g,i}} (u_i - \Delta P_{g,i} - \frac{1}{R_i} \Delta f_i) \\ \Delta \dot{P}_{t,i} = \frac{1}{T_{t,i}} \Delta P_{g,i} - \frac{1}{T_{t,i}} \Delta P_{t,i} \end{cases} \quad (2)$$

In these differential equations, $T_{t,i}$, the turbine time constant, $T_{g,i}$, the governor time constant, R_i , governor speed coefficient, D_i , load damping coefficient, M_i , the generator inertia constant of the i^{th} area, and T_{ij} synchronizing coefficient are. $\Delta P_{tie,i}$ and Δf_i are changes of tie-line power and frequency of the i^{th} area, respectively. $\Delta P_{g,i}$, $\Delta P_{t,i}$, and $\Delta P_{L,i}$ show the i^{th} area output power variations associated with the governor, turbine, and load, respectively.

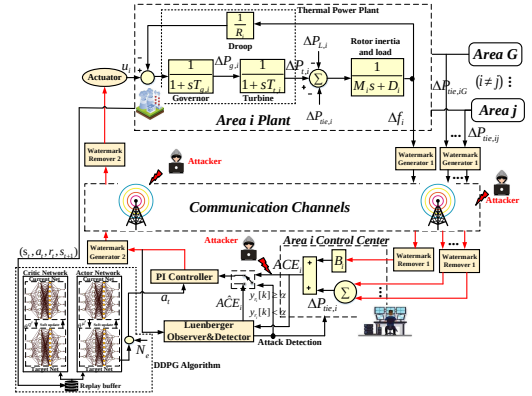


Fig. 1. Block diagram of the i^{th} control area for a G -area LFC system

III. DESIGN OF COVERT ATTACKS ON THE LFC SYSTEM

Fig. 2 depicts the description of the covert attack on the cyber-physical system. According to this figure, the attack signal $u_a(t)$ is injected into the watermarked control signal to disrupt the operation of the system, whereas the attack signal $y_a(t)$ is injected into the watermarked output signals to remove the impact of $u_a(t)$ on the measurement signal. The modeling of covert attacks can be represented by:

$$\begin{cases} \tilde{u}(t) = u(t) + u_a(t) \\ \tilde{y}(t) = y(t) - y_a(t) \end{cases} \quad (3)$$

where $\tilde{u}(t)$ and $\tilde{y}(t)$ are the control signal submitted to the physical system and measurement output signal obtained by the control center, respectively. The main hardship in identifying such attacks is that the attacker can design $u_a(t)$ and $y_a(t)$ to make the attack impact covert and not different from the expected behavior. The attacker aims to configure the disruption $u_a(t)$ such that the plant brings to some states determined by the attacker. Simultaneously, the effect of the attack $u_a(t)$ on the watermarked output signals y_g is canceled by $y_a(t)$. To this end, the attacker uses the dynamic system given by (4):

$$\begin{cases} \dot{\tilde{x}}_i(t) = A_i \tilde{x}_i(t) + B_i u_a(t) \\ y_a(t) = C_i \tilde{x}_i(t) \end{cases} \quad (4)$$

where $\tilde{x}_i(t)$ shows the state vector of the dynamic system used by the attacker (Attacker model of the plant in Fig. 2). Since the disruption $u_a(t)$ applied by the attacker is arbitrary, its features are generally unclear to defenders. Accordingly, model (4) is satisfactory for defining a covert attacker. The state-space equations of the system (1) under covert attacks can be represented as follows:

$$\begin{cases} \dot{\tilde{x}}_i(t) = A_i \tilde{x}_i(t) + B_i(u_i(t) + u_{a,i}(t)) \\ \tilde{y}_i(t) = C_i \tilde{x}_i(t) - y_{a,i}(t) \end{cases} \quad (5)$$

In this model, $\tilde{x}(t)$ and $\tilde{y}_i(t)$ show the vector of the state and output variables related to system (1) under covert attack. $y_{a,i}(t)$ is generated by (4) to cancel the impact of the input attack $u_{a,i}(t)$ against measurement signal.

Supposing that the adversary has a thorough knowledge about the system (matrices A_i, B_i, C_i) and the attack begins at time $t_c \geq 0$, the following conditions will illustrate the covert attack's stealthiness [12]. 1) If matrix A is Hurwitz, the attacker can suitably design the $y_a(t)$ to produce the covert attack as $t \rightarrow 0$. 2) If the attacker determines $\tilde{x}_i(t_c) = 0$ then the attack is covert for all instants and A .

IV. DESIGN OF THE PROPOSED DEFENSE MECHANISM

Fig. 2 demonstrates the proposed defense strategy structure in a cyber-physical system compromised by covert attacks. According to this figure, the proposed defense mechanism comprises the dynamic multiplicative watermarking-based filters, a Luenberger observer, a detector $\mathcal{D}_i$, and a controller. The

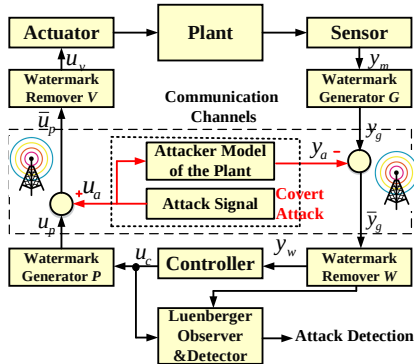


Fig. 2. Description of the covert attack on the cyber-physical system equipped with multiplicative watermarking technique

proposed multiplicative watermarking algorithm consisting of a watermark generator (filter before transmitting the original signal during the communication network) and a watermark remover (filter after transmitting the original signal during the communication network) is considered for measurement and control signals. Therefore, a watermark generator G along with a watermark remover W and a watermark generator P along with a watermark remover V are placed in the path of transmitting the measurement and control signals, respectively. It should be noted that the watermark generator and remover filters are the same for measurement signals. y_g, y_w, u_p , and u_v disclose the outputs related to the G, W, P , and V filters, respectively. $y_m, \bar{y}_g, u_c$, and $\bar{u}_p$ show the inputs of the G, W ,

P , and V filters, respectively. The watermarking filters and their equalizing are described as discrete-time linear systems with series interconnection as follows:

$$\begin{cases} x_\phi[k+1] = A_\phi x_\phi[k] + B_\phi \eta_\phi[k] \\ \lambda_\phi[k] = C_\phi x_\phi[k] + D_\phi \eta_\phi[k] \end{cases} \quad (6)$$

where $\phi \in \{g, w, p, v\}$ indicates whether the state, input, or output vectors are related to G, W, P , and V filters. x_ϕ, η_ϕ , and λ_ϕ show the vectors of the state, input, and output relevant to the watermark filters, respectively. $A_\phi, B_\phi, C_\phi, D_\phi$ are the parameters of the watermark generator and remover filters. These parameters must provide stability and invertibility between G and W , and P and V . The systems are stable and square:

$$\begin{cases} [\eta_g^\top, \eta_w^\top, \eta_p^\top, \eta_v^\top]^\top \doteq [y_m^\top, \bar{y}_g^\top, u_c^\top, \bar{u}_p^\top]^\top \\ [\lambda_g^\top, \lambda_w^\top, \lambda_p^\top, \lambda_v^\top]^\top \doteq [y_g^\top, y_w^\top, u_p^\top, u_v^\top]^\top \end{cases} \quad (7)$$

Moreover, all matrices have appropriate dimensions. The design of the G and P filters is such that they are stable inverses of W and V filters, respectively; hence $W \doteq G^{-1}$ and $V \doteq P^{-1}$ [17]. As a result, when there is no attack, we have:

$$\begin{aligned} WG = I &\rightarrow y_m[k] = y_w[k] \\ VP = I &\rightarrow u_c[k] = u_v[k] \end{aligned} \quad (8)$$

If an attack happens in the system, the Eq. (8) is violated. In the proposed strategy, the Luenberger observer presents the expected behavior of the plant by estimating the system's state variables. Hence, excessive deviation of the system output responses from the expected values indicates the happening of an attack on the system. To this end, the design of the Luenberger observer is accomplished as follows:

$$\begin{cases} \hat{x}_i[k+1] = A_{d_i} \hat{x}_i[k] + B_{d_i} u_i[k] + L_i y_{r_i}[k] \\ \hat{y}_{m_i}[k] = C_{d_i} \hat{x}_i[k] \\ y_{r_i}[k] = y_{w_i}[k] - \hat{y}_{m_i}[k] \end{cases} \quad (9)$$

In this formulation, $\hat{x}_i[k]$ and $\hat{y}_{m_i}[k]$ are the estimations $x_i[k]$ and $y_{m_i}[k]$ presented by the Luenberger observer at time step k , respectively. The $y_{r_i}[k]$ and L_i indicate the residual signal and the observer's gain, respectively. It is assumed that the $\hat{x}[0] = 0$. Also, the (A, C) pair should be a detectable pair. Considering the state estimation error $e[k] = x[k] - \hat{x}[k]$, the error derivative is calculated using $e[k+1] = (A - LC)e[k]$. The Luenberger gain must be specified so that all the eigenvalues of the matrix $(A - LC)$ are negative real values. According to Eq. (9), the Luenberger observer computes the residual signal using the W filter output. The residual signal becomes a significant value if an attack is imposed on the signals transmitted during the because the watermark remover filter W cannot reconstruct the original signal. Therefore, an imperfect signal is sent to the detector. After that, the detector $\mathcal{D}_i$ compares the residual signal $y_{r_i}[k]$ with a preset threshold value α to complete the attack detection process using the following equation:

$$\mathcal{D}_i : \begin{cases} y_{r_i}[k] \geq \alpha & , \text{ Occurrence of an attack} \\ y_{r_i}[k] < \alpha & , \text{ No attack} \end{cases} \quad (10)$$

α is chosen as the maximum residual signal value plus a security margin.

After the attack detection ($y_{r_i}[k] \geq \alpha$), the proposed detection strategy submits the estimated area control error (ACE) signal to the secondary controller instead of the measured ACE signal, as illustrated in Fig. 1. This paper employs a DRL-based PI controller as the secondary controller, ensuring frequency deviations are decreased under covert attacks. DRL algorithm employs a trained agent to train a machine to learn a specific task in an unknown dynamic environment. Based on a state-action mapping policy s_t , agents select the most efficient action (a_t) within the action space (A) for the current state of the environment ($\pi(a_t|s_t)$). The agent is given a reward r_t after every action, after which the environment is changed to a new state s_{t+1} after every action. For all states, this process is conducted again to maximize agents' expectations regarding discount rewards $R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$, where γ is a discount factor between 0 and 1 [5]. This study employs a deep deterministic policy gradient (DDPG) algorithm, which is an off-policy alternative among DRL algorithms. Searching for the optimal policy using DDPG strives to increase the anticipated cumulative long-term reward. DDPG uses two critic-actor deep neural networks with continuous state and action spaces for stable and robust learning. A replay buffer is also used to reduce the correlations of the input backgrounds. By mapping an environment state to a specific action, the actor-network of weight θ^μ sets the current policy $\mu(s_t|\theta^\mu)$. The critic network is $Q(s_t, a_t|\theta^Q)$, with a weight θ^Q . Stochastic gradient descent is used to update the critic network to minimize the loss function:

$$\begin{cases} L(\theta^Q) = E_{(s,a)}[(y_t - Q(s_t, a_t|\theta^Q))^2] \\ y_t = r_t(s_t, a_t) + \gamma Q(s_{t+1}, \mu(s_t|\theta^\mu)|\theta^Q) \end{cases} \quad (11)$$

In addition, the actor-network parameters are adjusted according to the following equation:

$$\nabla_{\theta^\mu} J^{\theta^\mu} \approx \mathbb{E}_{s_t \sim \rho^\beta} [\nabla_a Q(s, a|\theta^Q)|_{a=\mu^\theta(s)} \nabla_{\theta^\mu} \mu(s|\theta^\mu)] \quad (12)$$

where β indicates a specific policy to the current policy π , while ρ represents the state distribution discount. The target actor network and target critic network follow the trained networks utilizing Eq. (13):

$$\theta^{Q'} \leftarrow \nu \theta^Q + (1 - \nu) \theta^{Q'}, \theta^{\mu'} \leftarrow \nu \theta^\mu + (1 - \nu) \theta^{\mu'}, \nu \ll 1 \quad (13)$$

An exploration noise N_e is added to actor actions to ease the training performance through the suboptimal path neighborhood space. A reward as $r = -[|B_i \Delta f_i| + |\Delta P_{tie,i}|]$ is defined to ensure the DDPG algorithm has optimal performance.

V. SIMULATION RESULTS

The proposed scheme performance is compared with a conventional PI controller and an additive watermarking-based defense method, including the additive watermarking technique, a Luenberger observer, and a detector. The watermark signal considered in this paper is a Gaussian white noise with zero-mean and standard deviation $\sigma = 25 \times 10^{-4}$. More detail about this type of watermarking method is available in [18].

This paper neglects time delays as an assumption to provide a precise analysis of the proposed defense mechanism's effectiveness under covert attacks. The LFC system equipped with the developed control structure is implemented on the OPAL-RT device to prove the practicability of the presented scheme in real-time. Moreover, the root-mean-square error (RMSE) index ($\sqrt{\frac{1}{N_s} \times \sum_{k=1}^{N_s} ||h(k)||^2}$, where $h(k)$ and N_s show the frequency changes and the number of the simulation's samples, respectively,) and the index of the integral of time absolute error (ITAE) ($\int_0^t |ACE_i| dt$) are computed to numerically assess the performance of the considered defense methods. The threshold value α is equal to 0.004. The parameters of the studied LFC system are as: $T_{g1} = T_{g2} = 0.08s$, $T_{t1} = T_{t2} = 0.3s$, $T_{12} = 0.545p.u.$, $M_1 = M_2 = 0.167s$, $D_1 = D_2 = 0.0083p.u./Hz$, $R_1 = R_2 = 2.4Hz/p.u.$. To implement the covert attack on the system, the attacker injects the disruption u_a into the communication channels as pulse signals with the amplitudes of 0.1, 0.15, 0.18, and 0.2 at $15 < k \leq 25$, $35 < k \leq 45$, $55 < k \leq 65$, and $70 < k \leq 80$, respectively. Moreover, the load power changes in the range of $[0, 0.06]$ p.u.MW are randomly considered for each area. The attack signal y_a is produced using Eq. (4).

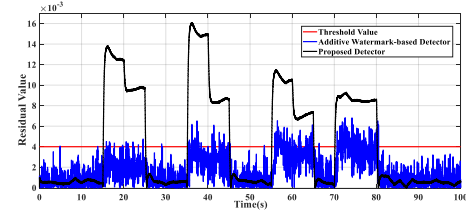


Fig. 3. Performance of the considered methods to detect covert attacks

Fig. 3 demonstrates the performance of the watermarking-based methods to detect covert attacks on the LFC system. As shown, the additive watermarking-based method cannot successfully detect the designed attacks. For example, the detector output (blue curve) at $15 < k \leq 25$ is below the threshold value (red line). Therefore, the attack is not detected during this time range. In contrast, the detector output of the proposed method has surpassed the threshold value's red line, revealing the attacks happening in the system. Accordingly, the proposed method successfully detects the designed attacks. In the additive watermarking-based methods, there is a possibility that the noise related to the Gaussian variable exceeds the threshold value, and the detector mistakenly identifies the noise as an attack. In addition, the additive watermarking-based approaches apply a more considerable computational burden to the system. Fig. 4 shows the output responses related to the watermark generator P and watermark remover V for control signals in the proposed method under covert attacks. According to these figures, the output of the watermark generator filter before and after the communication channel (P and $\sim P$) is different during attacks imposed on the communication channels. Accordingly, the multiplication of the watermarking filter and its equalizing filter (PV) is not 1 in the periods of the occurrence of the attacks. The dynamic responses obtained by the considered defense methods are

depicted in Figs. 5 (a)-(c). As shown, the proposed method presents much more suitable responses than the other strategies from viewpoints, decreasing the settling time and oscillations' amplitude under covert attacks. Also, the RMSE values of the frequency changes and the area's ITAE values for the considered methods are indicated in Table I. Based on this table, the proposed scheme presents much smaller RMSE and ITAE values than the other defense methods. Therefore, the proposed defense strategy has the capability of detecting and mitigating covert attacks.

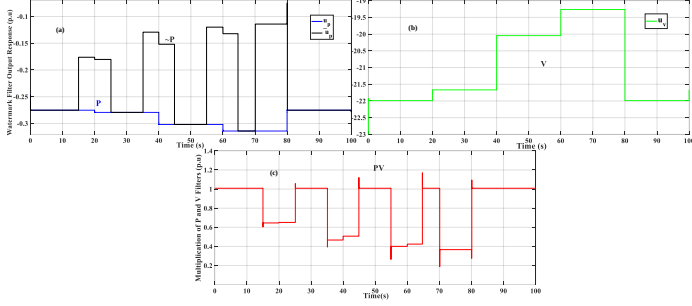


Fig. 4. Response of the watermark generator and remover filters P and V

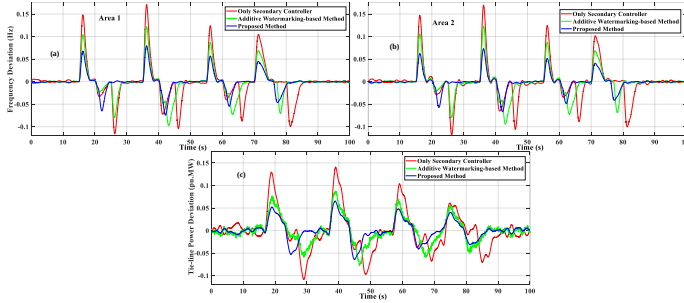


Fig. 5. Dynamic responses obtained by the considered defense methods

TABLE I
NUMERICAL ANALYSIS OF CONSIDERED DEFENSE METHODS

Method	RMSE		ITAE	
	Δf_1	Δf_2	Area1	Area2
Only Secondary Controller	0.0473	0.0468	0.362	0.331
Additive watermarking Method	0.0356	0.0364	0.257	0.223
Proposed Method	0.0239	0.0207	0.118	0.073

VI. CONCLUSION

This research investigated the LFC performance of a two-area power network against covert attacks. Accordingly, the combination of the Luenberger observer and multiplicative watermarking-based technique was proposed to detect covert attacks. A DDPG-based PI controller was also employed as the secondary controller to regulate the system's frequency. The real-time simulation results illustrated that the proposed strategy successfully copes with covert attacks imposed on the system and improves by the RMSE and ITAE indices at least 35% to other methods.

In future work, the proposed defense mechanism will be presented to cope with time delays, replay, and denial-of-service attacks.

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